RECEPTOR PHYSIOLOGY

Department of Medical Physiology and Microbiology

College of Medicine

Hawler Medical University

2025-2026

Learning Outcomes:

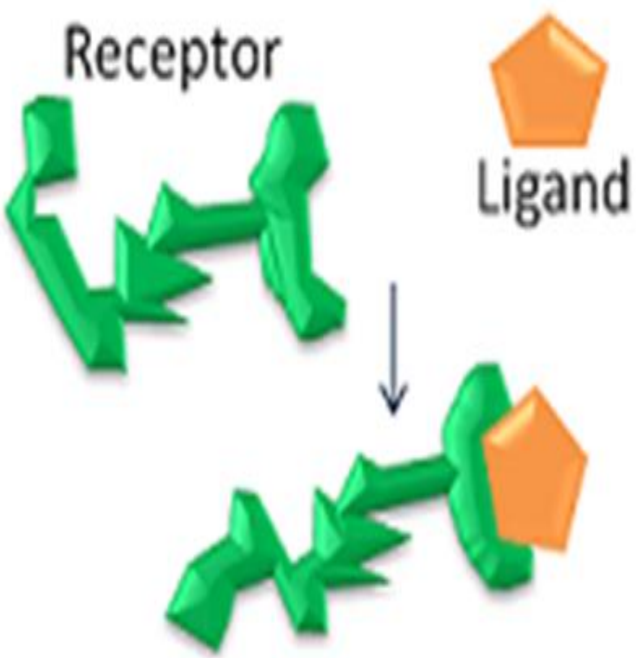
- Define receptors
- Classify the main types of receptors
- Categorize cell-surface receptors
- Identify Sensory receptors

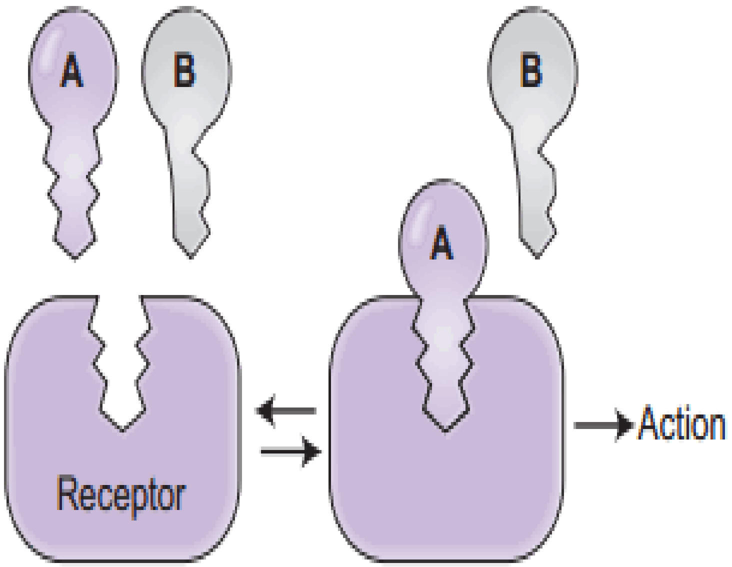
DEFINITION OF RECEPTOR

- A molecule (as a protein) on the cell membrane or inside a cell that selectively receives and binds a specific complementary molecule (ligand) such as hormones, neurotransmitters, antigens, or drug, to produce some effect in the target cell.

- In another words, receptor are protein molecule interacting with extracellular physiological signals and converting them into intracellular effects.

- Interaction of ligand with receptors is like a **lock and key.**

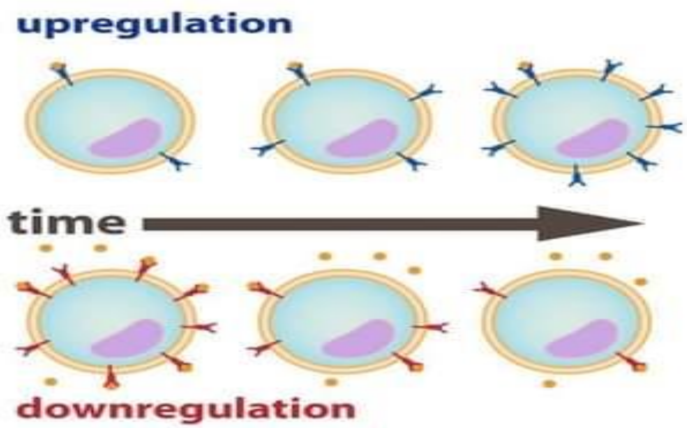




Drug **A** binds to receptor
Drug **B** cannot bind to receptor

- Number of receptors can change:

- **Down-regulation** - number of receptors decreases & target is less sensitive to a hormone or another agent. For example, insulin receptors may be downregulated in type 2 diabetes.



- **Up-regulation** - number increases & target is more sensitive. One example of up-regulation can be seen when people exercise more, increasing the sensitivity of cells to insulin.

CLASSIFICATION

- There are 2 types of receptors:

  Intracellular & cell surface receptor.

- **Intracellular (cytoplasmic receptors or nuclear receptors):**

 respond to hydrophobic ligand molecules
 Hydrophobic signaling molecules typically diffuse

across the plasma membrane.

- **Cell surface receptor /transmembrane receptors:**

- The signaling molecules (ligand) which is water-soluble substances, cannot diffuse across the plasma membrane; they bind to cell membrane receptors.

- performs signal transduction, converting an extracellular signal into an intracellular signal.

- There are three general categories of cell-surface receptors:

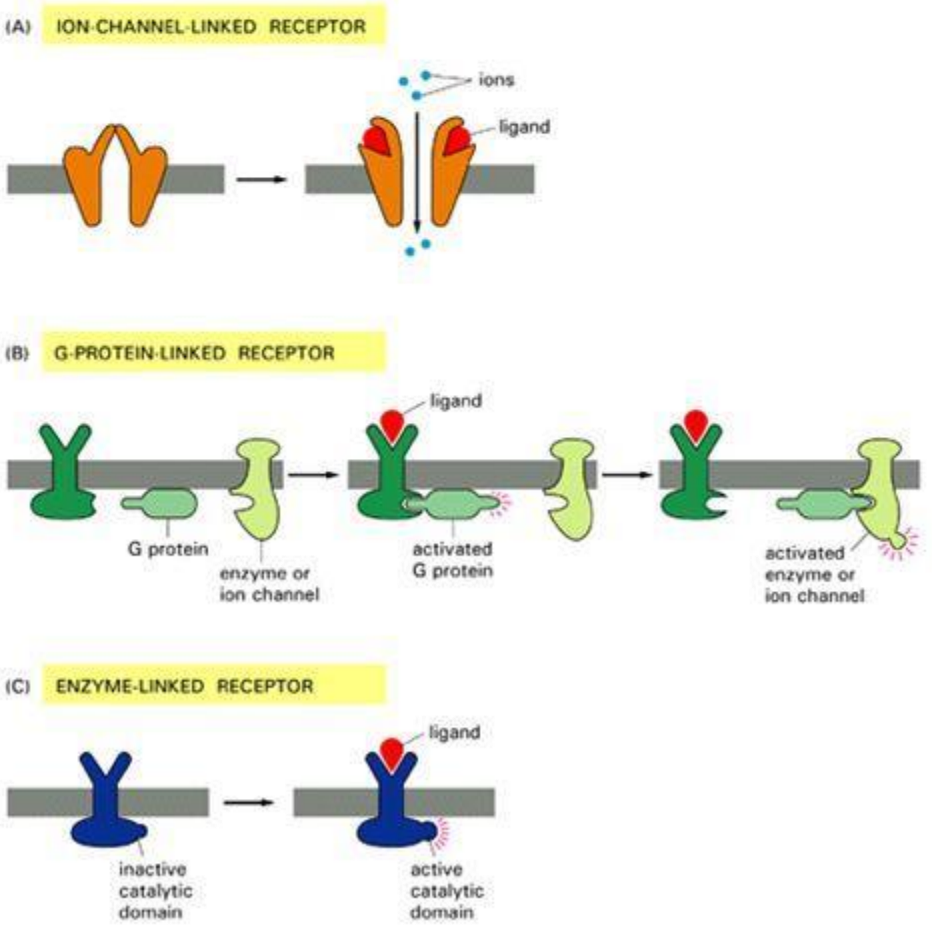
1. Ion channel-linked receptors

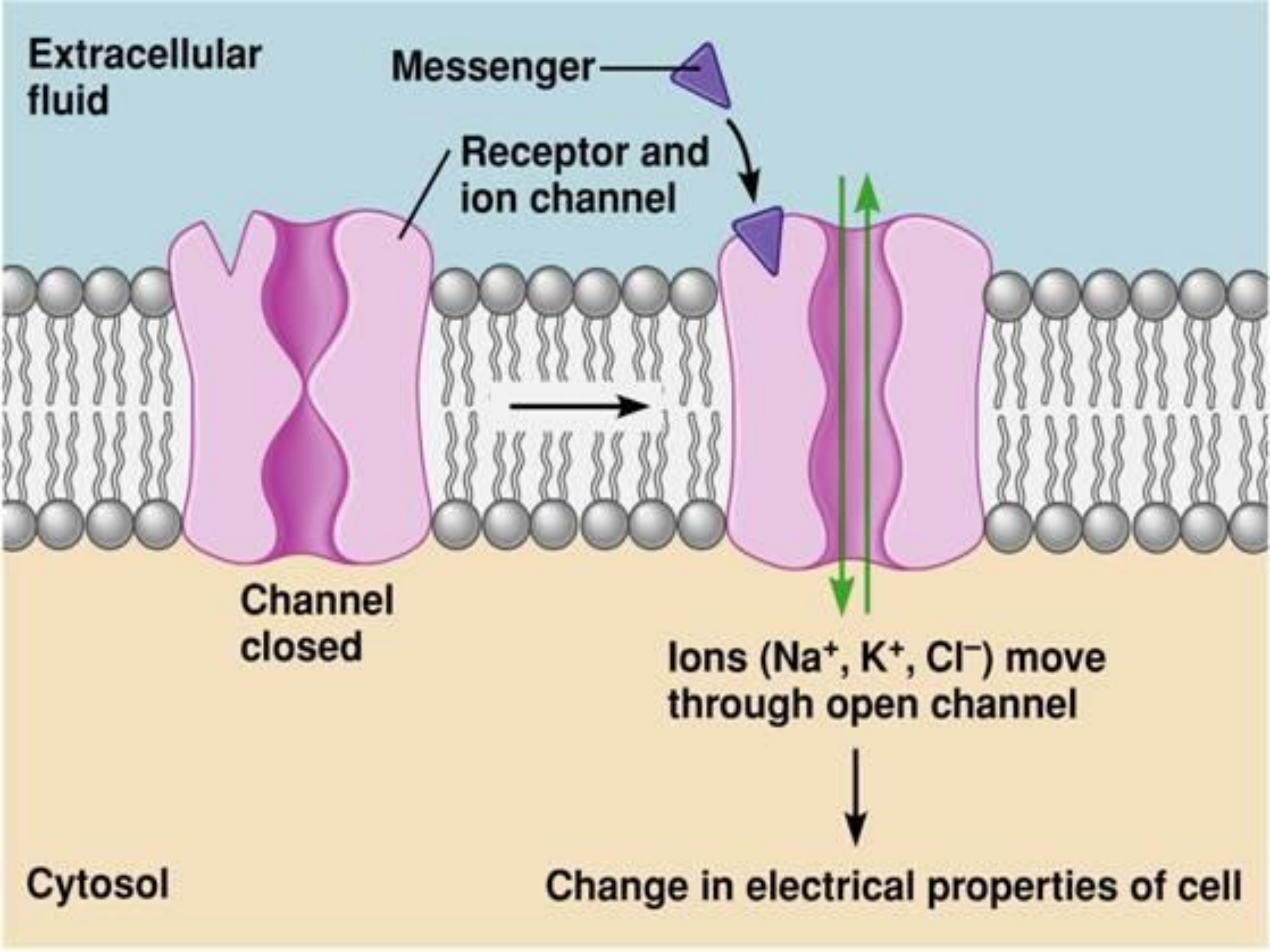
2. G-protein-linked receptors

3. Enzyme-linked receptors.

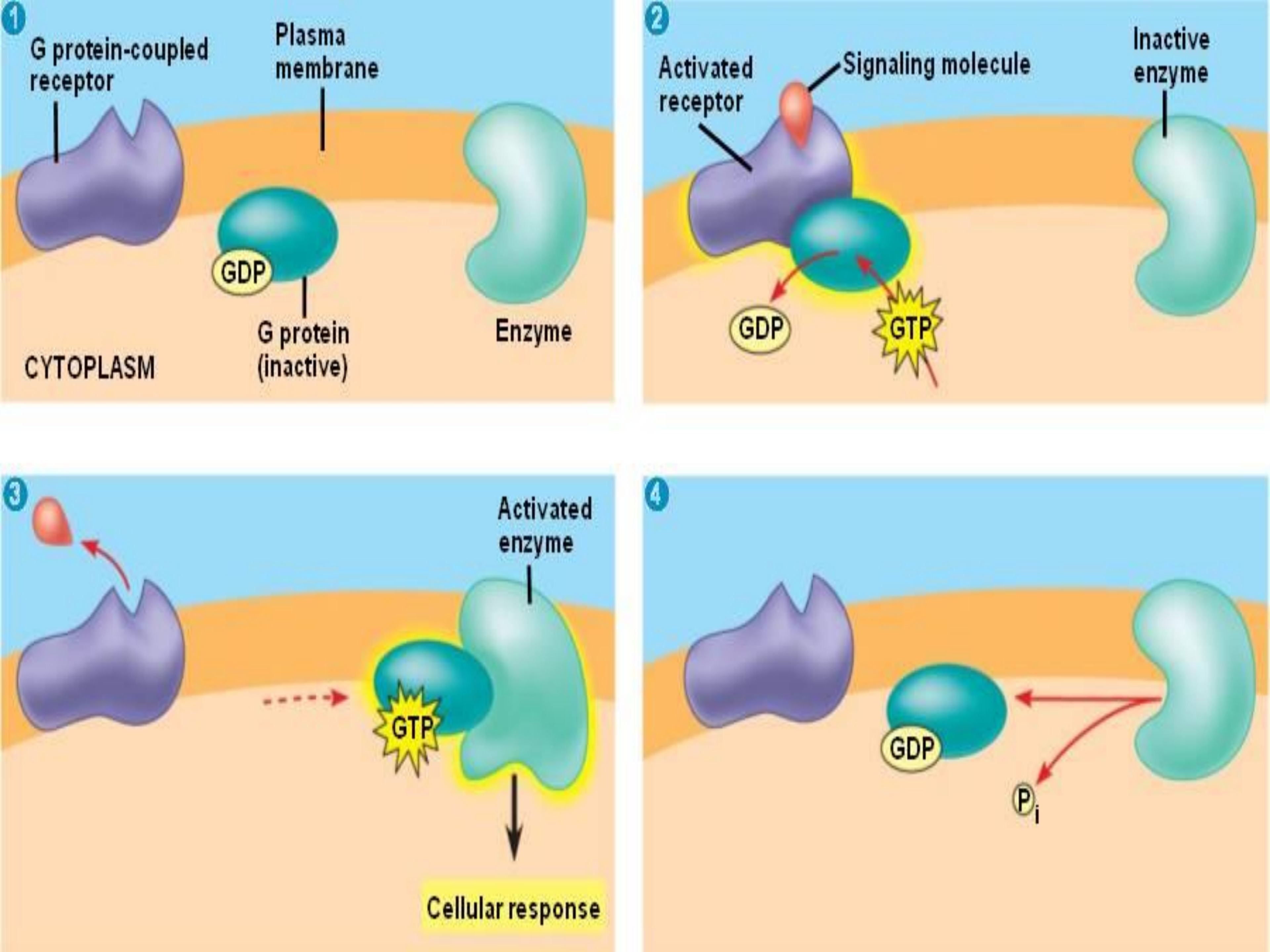
Cell Surface Receptors

- *Ion channel linked*: Binding of ligand causes channel to *open or close*
- *G-protein linked*: Binding of ligand *activates* a G-protein which will activate a separate enzyme or ion channel
- *Enzyme linked receptor*: Binding of ligand *activates an enzyme* domain on the receptor itself or on an associated molecule



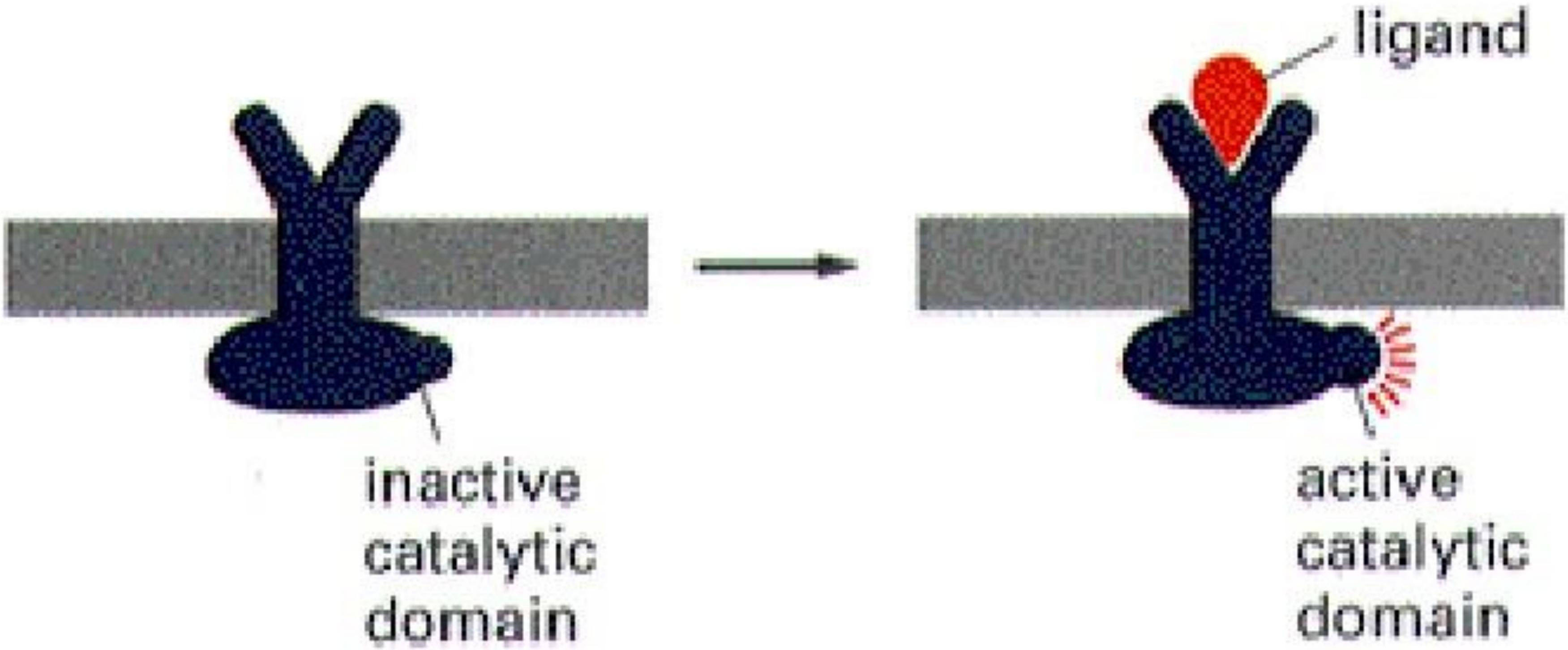


Extracellular fluid
Messenger
Receptor and ion channel
Channel closed
Ions (Na^+, K^+, Cl^-) move through open channel
Change in electrical properties of cell
Cytosol


1
G protein-coupled receptor
Plasma membrane
GDP
G protein (inactive)
Enzyme
CYTOPLASM
2
Activated receptor
Signaling molecule
Inactive enzyme
GDP
GTP
3
Activated enzyme
GTP
Cellular response
4
GDP
P_i

Cell Surface Receptor Types:

3) Enzyme-linked Receptor
eg Growth Factor Receptors



- **Ion Channel-Linked Receptors**

- Receptors bind with ligand (e.g. Nicotinic Receptor).
- Open a channel through the membrane that allows specific ions to pass through.
- Conformational change in the protein's structure that allows ions such as Na, Ca, Mg, and H2 to pass through.

- **G-Protein Linked Receptors**

- Binds with a ligand and activate a membrane protein called a G-protein.

- The activated G-protein then interacts with either an ion channel or an enzyme in the membrane.

- Each receptor has its own specific extracellular domain and G-protein-binding site. Example : Beta-adrenergic receptor.

- **Enzyme-Linked Receptors**

- Cell surface receptors with intracellular domains that are associated with an enzyme.

- Normally have large extracellular and intracellular domains.

- When a ligand binds to the extracellular domain, a signal is transferred through the membrane and activates the enzyme, which eventually leads to a response. e.g. : Tyrosine Kinase receptor.

SENSORY RECEPTOR

- A specialized cell or group of nerve endings that responds to sensory stimuli, like various sensory nerve endings in the skin, deep tissues, viscera, and special sense organs.

- Receptors are often the biological transducers, which convert (transducer) various forms of energy (stimuli) in the environment into action potentials in nerve fiber.

CLASSIFICATION OF SENSORY RECEPTORS

- Generally, sensory receptors are classified into two types:

A. Exteroceptors

B. Interoceptors

- **EXTEROCEPTORS**
  - Exteroceptors are the receptors, which give response to stimuli arising from outside the body.
- **INTEROCEPTORS**
  - Interceptors are the receptors, which give response to stimuli arising from within the body.

- **Exteroceptors** are divided into three groups:

1. Cutaneous Receptors or Mechanoreceptors

- Receptors situated in the skin such as touch, pain and pressure.

2. Chemoreceptors

- Receptors, which give response to chemical stimuli,

3. Telereceptors

- The receptors that give response to stimuli arising away from the body.

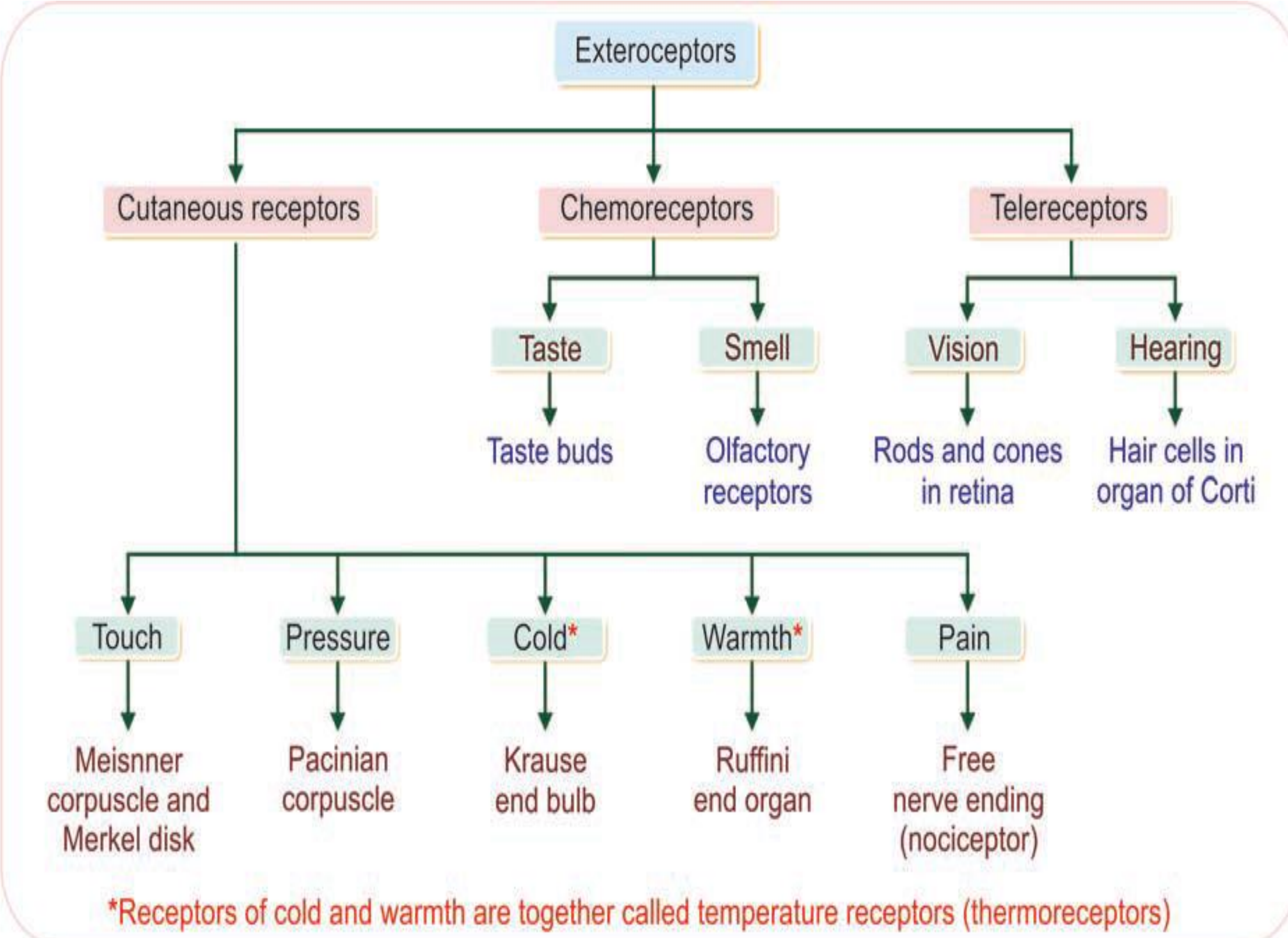
- **Interoceptors** are of two types:

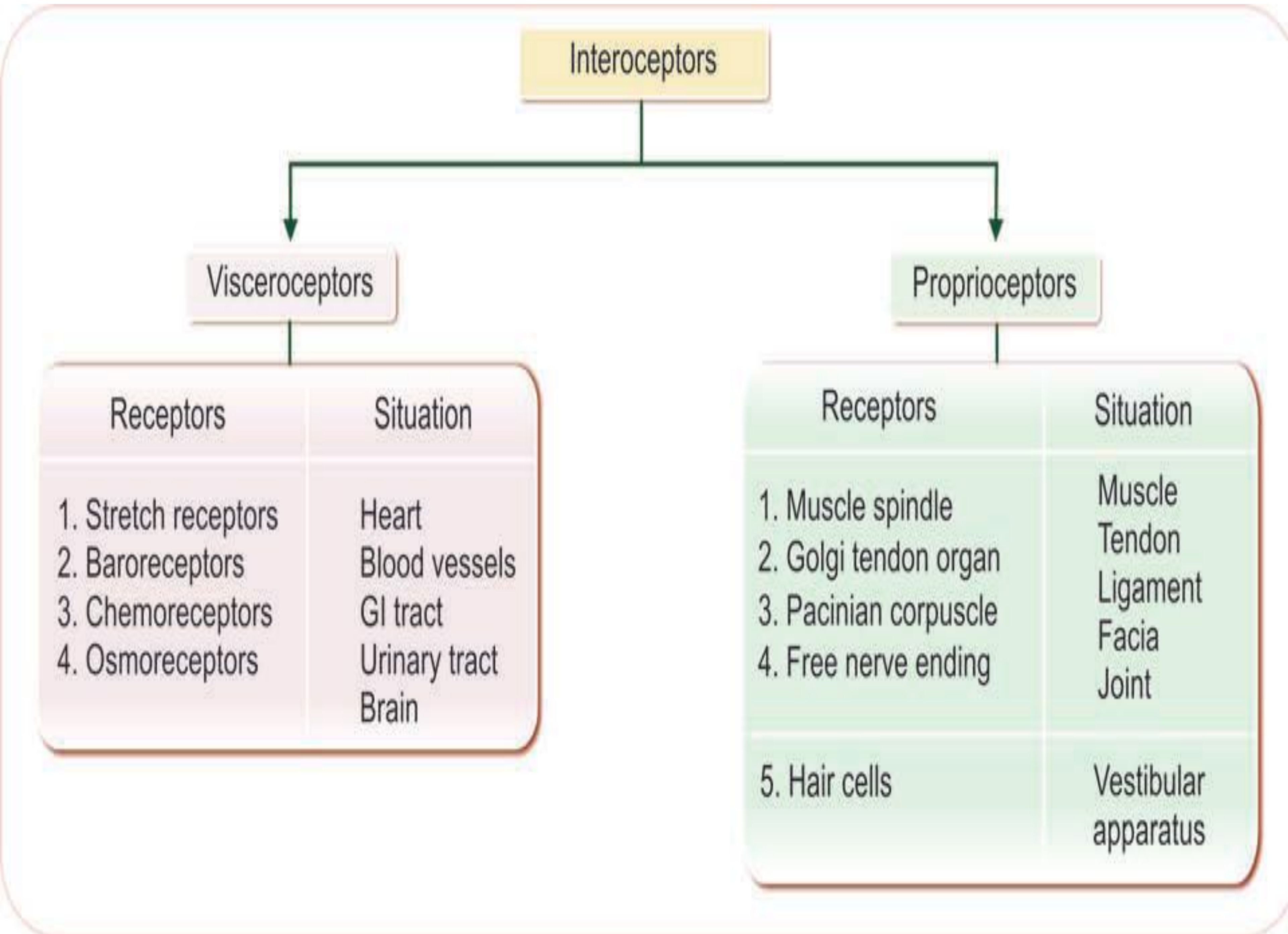
1. Visceroceptors

- Receptors situated in the viscera.

2. Proprioceptors

- The receptors, which give response to change in the position of different parts of the body.


Exteroceptors
Cutaneous receptors
Chemoreceptors
Telereceptors
Taste
Smell
Vision
Hearing
Taste buds
Olfactory receptors
Rods and cones in retina
Hair cells in organ of Corti
Touch
Pressure
Cold*
Warmth*
Pain
Meisnner corpuscle and Merkel disk
Pacinian corpuscle
Krause end bulb
Ruffini end organ
Free nerve ending (nociceptor)
*Receptors of cold and warmth are together called temperature receptors (thermoreceptors)


Interoceptors
Visceroceptors
Proprioceptors
Receptors
Situation
1. Stretch receptors
2. Baroreceptors
3. Chemoreceptors
4. Osmoreceptors
Heart
Blood vessels
GI tract
Urinary tract
Brain
Receptors
Situation
1. Muscle spindle
2. Golgi tendon organ
3. Pacinian corpuscle
4. Free nerve ending
Muscle
Tendon
Ligament
Facia
Joint
5. Hair cells
Vestibular
apparatus

THANK YOU